

Factorial Experiment Design

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Basic Factorial Design

The reason you conduct factorial experiments is to see how factors impact the response variable, as well as how each variable interacts with all combinations of the other variable. This is known as a full factorial experiment.

However, full factorial designs can be very costly and unnecessary when performing experiments on many factors. Statisticians use fractional factorial designs during early stages of experimentation for factor selection.

Sir Ronald A. Fisher wrote that complex designs, like factorial designs, are more efficient than studying each factor individually.

"Nature will best respond to a logical and carefully thought out questionnaire; indeed, if we ask her a single question, she will often refuse to answer until some other topic has been discussed."

Each factor will have 2 levels, high and low. For these notes, for some factor A, I will use A^+ for effect of A at high level.

And I will use A^- for effect of A at low level.

- Preliminary Product Process Design (PPPD) = factor screening
- After screening through potential inputs you can test more levels for each selected factor
- Need minimal experimentation
- Naive approach (randomly guessing) leads to way too many runs/trials

Steps for a Factorial Design

1. Determine the factors and levels for each factor
2. Set up way to process experimental results

k = number of factors
 l = number of levels for each factors
 A 2-Factorial design will require 2^k runs

Example

Suppose you are a farmer and you want to improve wheat production. You select 2 predictors, A = water and B = sunlight, both at 2 levels, high and low. You then collect data and observe the following:

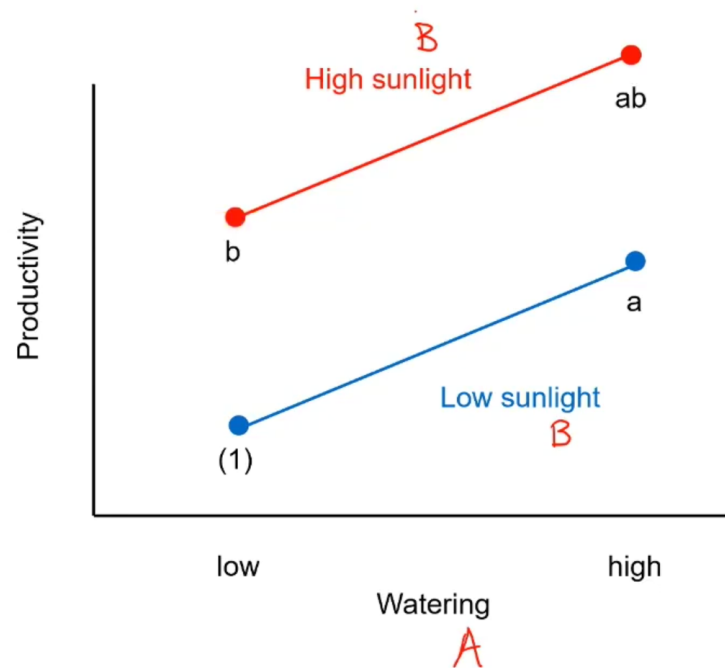


Figure 1: 2^2 Factorial Design Example

Interaction Effects

Interaction describes how the effect of one factor depends on the level of another factor. For example

Factor A = no caffeine, caffeine

Factor B = 4 hours of sleep, 8 hours of sleep

In english, when you sleep only 4 hours, caffeine improves reaction time a lot.

Sleep \ Caffeine	No Caffeine	Caffeine
4 hours	350 ms	300 ms
8 hours	250 ms	240 ms

Table 1: Reaction time (ms) by sleep duration and caffeine condition.

But when you sleep 8 hours caffeine only helps a little. So the benefit of caffeine changes depending on sleep level.

Graphically, interaction is shown by non-parallel lines. If the lines are almost parallel, then there is almost no interaction. If the lines are crossing, then there is significant interaction.

Here is an example showing a significant interaction.

Experience \ Method	Method A	Method B
Beginners	70	90
Advanced	95	80

Table 2: Example of a crossing interaction: the effect of teaching method reverses depending on experience.

Response Surface

In 2^2 factorial design you can model a response surface. The x-axis is factor A, the y-axis is factor B, and the z-axis (height) is the response you get at every (x,y). If this surface is a plane, it signals no interaction. If the surface is folded, then interaction is present.

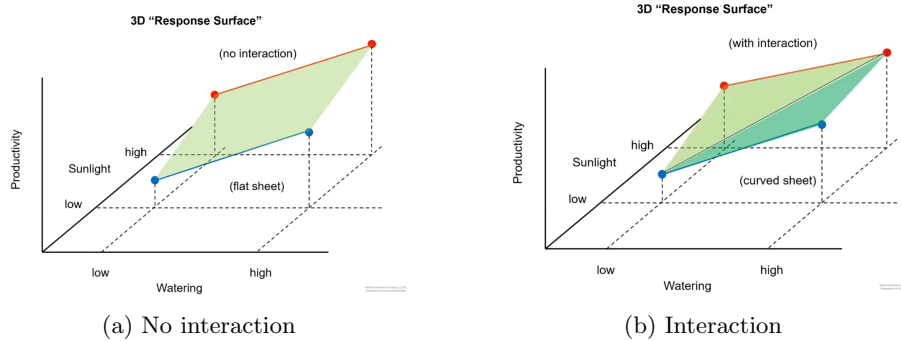


Figure 2: Response surface comparison

Effect Levels

Factors may be quantitative or qualitative (categorical).

A common notation for labeling different treatments (effect combinations) is the following:

Response (y)	x_1	x_2
(1)	A^-	B^-
a	A^+	B^-
b	A^-	B^+
ab	A^+	B^+

Table 3: Treatment levels for 2^2 factorial design

A more common way of coding these variable is using -1 for low levels and 1 for high levels.

Response (y)	x_1	x_2
(1)	-1	-1
a	1	-1
b	-1	1
ab	1	1

Table 4: Treatment levels for 2^2 factorial design

Effect of A and B

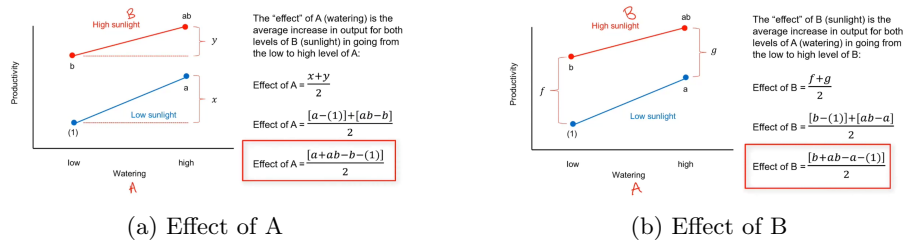


Figure 3: Response surface comparison

Replication

A replicate is a repeated trial performed under the same treatment conditions to measure natural variation (error).

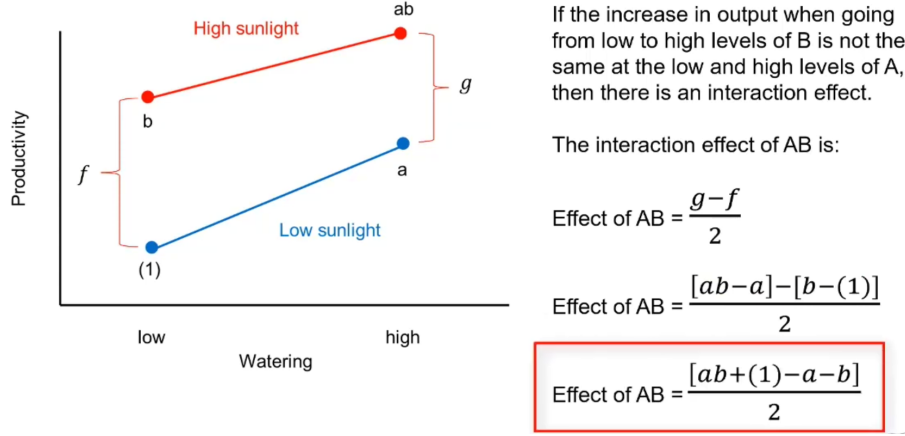


Figure 4: Effect of AB

$$\text{Effect of A} = \frac{a + ab - b - (1)}{2n}$$

$$\text{Effect of B} = \frac{b + ab - a - (1)}{2n}$$

Where n is the number of replicates. This difference is that ab, a, b, and (1) now represent that sum of all of our observations at each treatment level. So dividing by n gives us the average.

Contrast

Contrast is the specific linear combination of treatment means. For example, above we see that the effect of A is $\frac{a+ab-b-(1)}{2}$ for 1 replicate. The contrast of A is:

$$L_A = \langle 1, 1, -1, -1 \rangle \cdot \langle a, ab, b, (1) \rangle$$

or

$$L_A = \mu_a + \mu_{ab} - \mu_b - \mu_{(1)}$$

L is the contrast which is a linear combination of coefficients (c'_i s) and treatment means (μ' s).

$$L = c_1\mu_1 + c_2\mu_2 + \dots + c_k\mu_k$$

Such that $\sum_{i=1}^k c_i = 0$